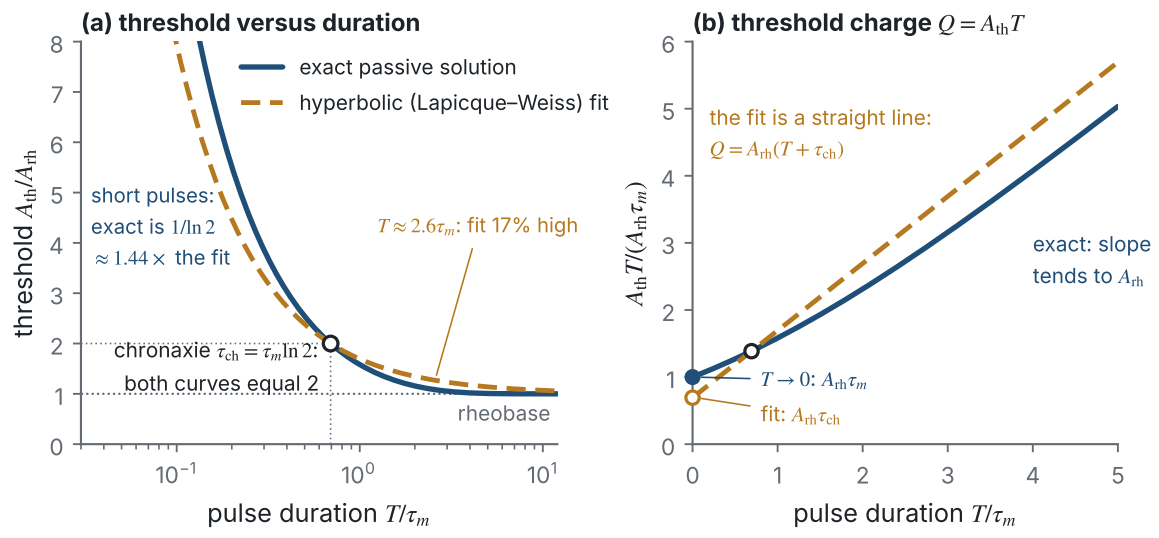
(a) threshold versus duration
exact passive solution
hyperbolic (Lapicque–Weiss) fit
short pulses:
exact is $1/\ln 2$
$\approx 1.44\times$ the fit
$T \approx 2.6\tau_m$: fit 17% high
chronaxie $\tau_{ch} = \tau_m \ln 2$:
both curves equal 2
rheobase
threshold A_{th}/A_{rh}
pulse duration T/τ_m
(b) threshold charge $Q = A_{th}T$
the fit is a straight line:
$Q = A_{rh}(T + \tau_{ch})$
exact: slope
tends to A_{rh}
$T \to 0$: $A_{rh}\tau_m$
fit: $A_{rh}\tau_{ch}$
$A_{th}T/(A_{rh}\tau_m)$
pulse duration T/τ_m